INCH-POUND
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PERFORMANCE SPECIFICATION
MECHANICAL SCANNER, MX-9872(V)/UA;
GENERAL SPECIFICATION FOR

This performance specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers the general attributes of the common module, Mechanical Scanner, MX-9872(V)/UA. This specific coverage of each variation of the MX-9872(V)/UA are contained in specification sheets attached as supplements to this specification.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification whether they are listed or not.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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2.2 Government documents.

2.2.1 Specifications, Standards, and Handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and therefore are not mandatory.

STANDARDS

MILITARY

MIL-STD-810 Environmental Test Methods and Engineering Guidelines

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454 - Standard general requirements for electronic equipment.

(See supplement 1 for list of specification sheets.)

(Unless otherwise indicated, copies of Federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

USA CECOM

SM-D-807690 Scanner, Mechanical, MX-9872(v)/UA

(Copies of documents, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless

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otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IPC-S-815A	- General requirements for soldering electronic inter-connections.

(Application for copies should be addressed to The American National Standards Institute, Inc., 1430 Broadway, N.Y., N.Y. 10018)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E122-89	- Sample size to estimate a quantity for a lot or process.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103-1187)

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQC Z 1.4	-Sampling Procedures and Tables for Inspection by Attributes.

(Applications for copies should be addressed to American Society for Quality Control, PO Box 3005, 611 East Wisconsin Avenue, Milwaukee, Wisconsin 53202)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification Sheet. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Materials. The contractor shall select the materials, but the materials and parts used

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shall be fully capable of meeting all of the operational and environmental requirements specified herein. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard.

3.4 Description. The Mechanical Scanner, MX-9872(V)/UA, herein known as the Scanner, normally consists of a scan mechanism assembly and housing. The scan mechanism includes a flat scan mirror, interlace gimbal, and associated drive components. The scan mirror pivots about the scan axis and oscillate about the interlace axis. The Scanner has provisions for mounting a phase lens and counterweight for system unique requirements.

3.4.1 Construction. The Scanner shall be constructed in accordance with this specification and, shall assure compliance with the contract configuration baseline and the form, fit and function interface requirements as specified in the applicable specification sheet. .

3.4.2 Weight. (See applicable specification sheet.)

3.5 Components. The Scanner consists of the following major components:

a. Scan mirror assembly
b. Solenoid (2)
c. Interlace flex pivot assembly (2)
d. Scan transducer assembly
e. Interlace transducer assembly
f. Upper scan flex pivot assembly
g. Lower scan flex pivot assembly
h. Return spring assembly (4)
I. Scanner housing
j. Scanner housing cover
k. Gimbal
l. Wiring harness
m. Gimbal counterweight

3.6 Performance characteristics. (See 3.1)

3.6.1 Scan angle and frequency.

3.6.1.1 Variable scan frequency. Scan frequency of the scanner shall be variable from 20 IR frames/40 fields to 62 IR frames/124 fields per second. For variable scan frequency measurement and verification, see 4.5.1.1.

3.6.1.2 Specific scan frequency (60 Hz). (See 3.1)

3.6.1.3 Synchronous scan frequency. (See 3.1)

3.6.1.4 Scan angle. (See 3.1) The scan angle shall be determined as follows:

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a. The test apparatus shall consist of a holding fixture, a projection system capable of projecting a spot or marker into the scanner along the x-axis (Figure 4) and an appropriate scale graduated in degrees measured about the scan mirror axis with its zero position located on the y-axis (Figure 4).

b. The Scanner shall be operated and the angular excursions of the projected spot in each direction shall be recorded. One-half of the sum of the excursions shall be the measured scan angle.

3.6.2 Interlace. The Scanner shall operate in a 2:1 or 1:1 interlace mode and shall interlace either once per azimuth scan or once every other azimuth scan, respectively. The interlace switching time defined in Figure 1 shall be as specified in the applicable specification sheet (see 3.1). Verify specified performance through the use of 4.5.2.

3.6.3 Scan direction. The Scanner shall operate in a bi-directional mode at 2 scans per cycle and shall display the waveform shapes and phase relationships as specified in the applicable specification sheet (see 3.1). Verify specified performance through the use of 4.5.3.

3.6.4 Power consumption. Power required to operate the Scanner and the Scan-Interlace module is dependent upon the specific Scanner configuration but shall not exceed 7.5 watts (See 3.1) Verify specified performance through the use of 4.5.4.

3.6.5 Scan efficiency. The scanner scan efficiency shall be as noted in the applicable specification sheet (see 3.1). Verify specified performance through the use of 4.5.5.

3.6.6 Scan jitter in azimuth. The amount of jitter measured in the scan direction shall not be greater than 0.75 milliradian as determined by using an oscilloscope. Verification of specified performance shall be demonstrated though use of 4.5.6.

3.6.7 Scan jitter in interlace. The interlace amplitude shall be repeatable to within a ±10% deviation of the height of a display element as measured over the center 80% of the azimuth field of view. Verify specified performance through the use of 4.5.7.

3.6.8 Modulation transfer function (MTF). The MTF of each side of the scan shall be no less than 95 % at 0.667 cycles per milliradian (cy/mrad). Verify specified performance through the use of 4.5.8.

3.7 Environmental conditions.

3.7.1 Temperature shock. The scanner shall not be damaged (see 3.14.1) by sudden changes in temperature between -54° and +95°C .

3.7.2 High temperature. The Scanner shall not be damaged (see 3.14.1) by storage to +95°C or continuous operation to +71°C, and shall operate for 30 minutes at +95°C. The

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average interlace waveform amplitude (Figure 1) shall not change more than ±20 percent from the initial setup at ambient temperature (see 3.14.3) over the specified high temperature operating range

3.7.3 Low temperature. The Scanner shall not be damaged (see 3.14.1) by operation to -54°C or storage to -62°C. After soaking at -62°C, the average interlace waveform amplitude (Figure 1) shall not change more than ±20 percent from initial setup at ambient temperature (see 3.14.3) when stabilized at -30°C .

3.7.4 Shock. The Scanner shall not be damaged (see 3.14.1) by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests. The average interlace waveform amplitude (Figure 1) shall return to the initial amplitude within 0.2 second after the Scanner is subjected to the operating shock environment, not to include bench handling test .

3.7.5 Vibration. The Scanner shall not be damaged (see 3.14.1) by vibration over a frequency spectrum at the specified g levels and amplitudes as shown in Figure 3 for the x, y, and z axes defined in Figure 4. During specified vibration, the average interlace waveform amplitude (Figure 1) shall not deviate from the amplitude without vibration by more than ±50 percent. The sinusoidal cycle time shall be 120 minutes per axis The dwell time shall be 1/6 of the cycling time at each resonance .

3.8 Soldering. Electrical soldering shall conform to ANSI/IPC-S-815A.

3.9 Reliability. The Scanner shall have a mean time between failure (MTBF) of least 6,058 hours $\theta_{\circ}$.

3.10 Nameplates and product marking. Unless otherwise specified, Scanner parts and components, subassemblies, and assemblies shall be marked for identification and reference designation markings in accordance with the applicable drawing or the requirements of the contract or purchase order. Use requirement 67 of MIL-HDBK-454 for guidance.

3.11 Treatment and painting. Unless otherwise specified, the Scanner and its components and parts shall have a surface treatment which will prevent rust, corrosion, and fungus growth in the full range of environmental requirements. External color and surface shine shall reduce light reflection and reduce recognition. Adherence must be assured through the full range of environmental and use factors.

3.12 Preconditioning. Each Scanner shall be operated for at least 8 hours prior to Group A inspection .

3.13 Workmanship. All electronic parts, components, assemblies or sub-assemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust, dirt or tool marks. Wires or integrated

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circuitry shall be protected from contact with rough or irregular surfaces and be shielded from shorting. MIL-HDBK-454 may be used as guidance.

3.14 Definitions.

3.14.1 Damage. Breakage, loosening, shifting, evidence of corrosion, failure of Any finish, hardware, connection or component; and any degradation of scanner performance to values less than specified herein.

3.14.2 Failure. A noted incapability to properly attain a specified performance requirement during the conduct of a test or examination process described within this specification.

3.14.3 Ambient temperature. Ambient temperature is +23°C ±5°C.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one or more complete Scanners when a first article sample is required (see 3.1). This inspection shall include the inspection of 4.2.1 and the tests of 4.2.2.

4.2.1 Inspections. All mechanical scanners shall be examined for conformance to the applicable specification or drawing. Inspections shall be made using Table I and sampling procedures per ANSI/ASQC Z1.4 and sample size per ASTM E122-89 and as directed by the contract or purchase order.

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TABLE I. Visual Inspections.

Inspection Requirement	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	3.5	3.5
Construction	3.4.1	See specification sheet
Weight not as specified	3.4.2	See specification sheet
Soldering not as specified	3.8	ANSI/IPC S-815A
Marking not as specified	3.10	See drawing or contract or purchase order.
Workmanship not as specified	3.13	3.13
Treatment and painting not as specified	3.11	See drawing or manufacturer's internal Procedure

4.2.2 Tests. Upon successful completion of inspections specified in 4.2.1, perform the tests of Table II in any order, except that the modulation transfer function (MTF) test shall follow the high temperature test. Failure of any tests shall constitute failure of first article inspection.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Variable scan frequency	3.6.1.1	4.5.1.1
Specific scan frequency	3.6.1.2	4.5.1.2
Synchronous scan frequency	3.6.1.3	4.5.1.3
Scan angle	3.6.1.4	4.5.1.4
Interlace	3.6.2	4.5.2
Scan direction	3.6.3	4.5.3
Power consumption	3.6.4	4.5.4
Scan efficiency	3.6.5	4.5.5
Scan jitter in azimuth	3.6.6	4.5.6
Scan jitter in interlace	3.6.7	4.5.7
Modulation transfer function	3.6.8	4.5.8
Temperature shock	3.7.1	4.6.1
High temperature	3.7.2	4.6.2
Low temperature	3.7.3	4.6.3
Shock	3.7.4	4.6.4
Vibration	3.7.5	4.6.5
Reliability	3.9	4.7

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4.2.3 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities .

4.3 Conformance inspection.

4.3.1 Inspection. Visual inspection shall be as specified in 4.2.1.

4.3.2 Tests.

4.3.2.1 Group A inspection. Group A inspection shall be conducted at ambient temperature on all scanners following the preconditioning of 3.12 and the visual inspection of 4.3.1. Group A tests listed in Table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Specific scan frequency	3.6.1.2	4.5.1.2
Synchronous scan frequency	3.6.1.3	4.5.1.3
Scan angle	3.6.1.4	4.5.1.4
Interlace	3.6.2	4.5.2
Power consumption	3.6.4	4.5.4
Scan efficiency	3.6.5	4.5.5
Scan jitter in azimuth	3.6.6	4.5.6
Scan jitter in interlace	3.6.7	4.5.7

4.3.2.2 Group B inspection. Not required.

4.3.2.3 Group C inspection. Group C inspections shall be conducted on scanners selected from units which have passed the tests in 4.3.2.1. The sample (s) shall be tested in accordance with the inspections listed in Table IV . Group C tests may be performed in any order, except that the modulation transfer function test 4.5.8 et al shall be performed last.

TABLE IV. Group C Inspection.

Inspection	Requirement paragraph	Test paragraph
Scan direction	3.6.3	4.5.3
High temperature	3.7.2	4.6.2
Low temperature	3.7.3	4.6.3
Shock	3.7.4	4.6.4
Vibration	3.7.5	4.6.5
Modulation transfer function	3.6.8	4.5.8

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4.3.2.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract (see 6.2).

4.3.2.3.3 Disposition of Group C samples. Group C samples shall be accepted on contract subsequent to the tests of Table III.

4.3.2.4 Group D inspection. Not required.

4.4 Sampling. Sampling for Group A and C shall be in accordance with the contract or purchase order.

4.5 Methods of inspection. Unless otherwise specified, the tests shall be performed utilizing the Scan-interlace module as specified in the applicable specification sheet.

4.5.1 Scan angle and frequency.

4.5.1.1 Variable scan frequency. The scanner shall be connected to Scan-Interlace, PL-1408/UA and mounted on a suitable fixture. Power shall be applied to the scanner and the output of the azimuth transducer (Scan-Interlace) shall be monitored with a calibrated frequency counter. The output of the scan drive electronics shall be varied to demonstrate that the requirements of 3.6.1.1 are met. Failure to meet the requirements of 3.6.1.1 shall constitute failure of this test.

4.5.1.2 Specific scan frequency. See applicable specification sheet.

4.5.1.3 Synchronous scan frequency. See applicable specification sheet.

4.5.1.4 Scan angle. The angle shall be determined as follows:

4.5.1.4.1 Test apparatus. The test apparatus shall consist of a holding fixture, a projection system capable of projecting a spot or marker into the scanner along the x-axis (Figure 4) and an appropriate scale graduated in degrees measured about the scan mirror axis with its zero position located on the y-axis (Figure 4).

4.5.1.4.2 Test conduction. The scanner shall be operated and the angular excursions of the projected spot in each direction shall be recorded. One-half of the sum of the excursions shall be the measured scan angle.

4.5.1.4.3 Failure criteria. Failure to meet the requirements of 3.6.1.4 shall constitute failure of this test.

4.5.2 Interlace. With the scanner operating in a suitable fixture, the output of the interlace position transducer shall be monitored with an oscilloscope. The waveform switching times

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(Figure 1) shall be measured to verify the requirement of 3.6.2 is met. Failure to meet requirements of 3.6.2 shall constitute failure of this test.

4.5.3 Scan direction. With the scanner operating, the outputs of the azimuth and interlace position transducers or the interlace transducer and scan tachometer (using the appropriate scan-interlace module) shall be observed with an oscilloscope. The output waveforms shall show the shape and phase relationships as specified in the applicable specification sheets. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.5.4 Power consumption. The scan drive electronics shall be connected to constant voltage power supplies. The current being drawn by the electronics from each voltage source shall be measured to obtain the average current being drawn. The power delivered to the module shall be calculated as the sum of the products of the power supply voltages and their average currents. Failure to meet requirements of 3.6.4 shall constitute failure of this test.

4.5.5 Scan efficiency. With the scanner operating, the period of the azimuth position transducer waveform shall be measured with an oscilloscope. The scanner shall then be stopped, and the scan mirror moved from contact with one return spring to contact with the opposite return spring. Simultaneously, the signal level of the transducer at the spring contact points shall be noted on the oscilloscope. With the scanner operating, the period of time that the transducer output is between the two signal levels during one cycle shall be determined from the oscilloscope display. The ratio of this period of time to the azimuth position transducer waveform period which was measured previously is the duty cycle or scan efficiency of the scanner operated with the appropriate scan-interlace module. See applicable specification sheet (see 3.1) when a scan-interlace module other than PL-1408/UA is used. Failure to meet the requirements of 3.6.5 shall constitute failure of this test.

4.5.6 Scan jitter in azimuth. The scanner shall be set up and operating as in 4.5.1.2. The scan jitter may be measured by use of an oscilloscope or similar method that will provide conclusive evidence that the scanner meets its requirements. Using an oscilloscope, the scan position zero crossing detection circuit shall be measured for jitter on any positive transition edge (see Figure 2). Failure to meet requirements of 3.6.6 shall constitute failure of this test.

4.5.7 Scan jitter in interlace. The output of the interlace position transducer shall be monitored with an oscilloscope. The average interlace movement signal, defined as the average of the difference between the waveform amplitude on the forward scan and the waveform amplitude on the reverse scan at identical scan mirror azimuth positions, shall be measured. Scan jitter in interlace is defined as the variation of the above differences from the average interlace movement signal. Failure to meet requirements of 3.6.7 shall constitute failure of this test.

4.5.8 Modulation transfer function (MTF). The scanner shall be located in a 38.0, ±0.5 mm collimated bundle of visible radiation (660 nanometers peak, with 20 nanometer half-power bandwidth) parallel and centered on the optical axis. The scanner shall be tested on the MTF apparatus that yields sine wave response. The image area to be analyzed shall be at least ±1.5

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milliradians around the optical axis. The degradation due to the scanner shall be determined by comparison of the MTF measurements with the test apparatus MTF.

4.5.8.1 Test apparatus MTF. The MTF for the test apparatus shall be determined by rotating the optical imaging system to be aligned to the collimated beam. The maximum MTF of the test apparatus shall be determined by an optimization search of the image.

4.5.8.2 MTF. The MTF of the scanner shall be measured in both azimuth and elevation directions for each mirror surface and there shall be no refocusing allowed between the azimuth and elevation requirements. In each direction and for each side, the ratio of the scanner MTF at 0.667 cycles/milliradian to the corresponding MTF measurement of the test apparatus (see 4.5.8.1) shall be obtained. Failure to meet the requirement of 3.6.8 shall constitute failure of this test.

4.6 Environmental tests. The environmental tests (4.6.1 through 4.6.4) shall be performed in accordance with MIL-STD-810. Vibration tests will be performed in accordance with the Test Method specified in Appendix A (see 4.6.5). The testing required during these tests shall be as specified below. All performance tests specified shall be performed within 5 minutes of completion of environmental tests.

4.6.1 Temperature shock. The scanner shall be tested in accordance with MIL-STD-810, Test Method 503.3, Section II, Procedure II-3, except that the temperature of step 1 and 3 shall be raised to +95°C. The temperature shall be maintained at +95°C for a minimum of 4 hours until the item has stabilized. The temperature of step 4 shall be lowered to -54°C and shall be maintained at this temperature for a minimum of 4 hours until the item has stabilized. The scanner shall not be operated during temperature shock tests (no measurements shall be taken during tests). Failure to meet requirements of 3.7.1 shall constitute failure of this test.

4.6.2 High temperature. The scanner shall be tested in accordance with MIL-STD-810, Test Method 501.3, Section II-3: Procedure I (storage): the temperature of step 4 shall be raised to +95°C and maintained for a minimum of 48 hours. The scanner shall be operated during the final 30 minutes of step 4 (no measurements shall be taken during step 4). Procedure II (operation): the temperature of step 2 shall be +71°C (use Constant Temperature Exposure). The scanner shall be operated at +71°C. The high temperature test criteria specified in 4.5.1.2 and the average interlace waveform amplitude shall be measured in place of the table III tests during high temperature tests. Failure to meet requirements of 3.7.2 shall constitute failure of this test.

4.6.3 Low temperature. The scanner shall be tested in accordance with MIL-STD-810, Test Method 502.3, Section II-3: Procedure I (storage): the temperature of step 2 shall be -62°C and shall be maintained for a minimum of 48 hours. Procedure II (operation): the temperature of step 1 shall be -54°C. The low operating temperature for the interlace waveform criterion shall be -30°C. The low operating test criteria specified in 4.5.1.2 and the average interlace waveform amplitude shall be measured in place of the Table III tests during low temperature tests. Failure to meet requirements of 3.7.3 shall constitute failure of this test.

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4.6.4 Shock. The scanner shall be tested in accordance with MIL-STD-810, Test Method 516.4, Section I-3: Procedure I (Functional Shock – operation): the scanner shall be subject to high intensity shock per Figure 516.4-4 with a peak acceleration magnitude of 100 g's at 11 milliseconds time duration. During this procedure, the scanner shall be operated and the average interface waveform amplitude shall be measured in lieu of Table III. Procedure VI (Bench Handling): the scanner shall be subject to the Bench Handling tests of Procedure VI. Failure to meet requirements of 3.7.4 shall constitute failure of this test.

4.6.5 Vibration. The scanner shall be tested in accordance with Test Method 514.2 (see Appendix A), Procedure I (category c, Table 514.2-III, Fig. 514.2-3) , and, Procedure VIII (category f, Table 514.2-VI, Figure 514.2-6), except that the curve shown in Figure 3 of this specification shall be used in lieu of the curve of Appendix A. The sinusoidal cycle time shall be 120 minutes per axis. The dwell time shall be 1/6 of the cycling time at each resonance. The scanner shall be operated during vibration and the average interlace waveform amplitude shall be measured in place of the Table III test. Failure to meet requirements of 3.7.5 shall constitute failure of this test.

4.7 Reliability.

4.7.1 First article reliability. The lower test MTBF (θ) shall be demonstrated in accordance with the following subparagraphs:

4.7.1.1 Test level.

a. Temperature +50°C, (+5 -0)°C

b. Temperature cycling - None

c. Vibration - 2.2 g, ±10 percent peak acceleration value at any nonresonant frequency between 20 and 60 Hz. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The vibration shall be in the direction of the x-axis (see Figure 3).

d. Power on-off cycling - Turn power on and let temperature stabilize, hold for 3 hours, then turn power off and let temperature stabilize. This cycle shall continue throughout the test.

4.7.1.2 Parameters. The discrimination ratio θ_o/θ_1 shall be 3.

4.7.1.3 Failure criteria and test length. The tests of Table III shall be performed at least once prior to and once after the tests of 4.7.1.1. Additional testing of Table III may be selected by the contractor. The results of all Table III tests and any physical damage shall be used to determine the success or failure of the reliability test. In the event of a Table III test failure, the time interval between the preceding successful test and the failure shall not contribute to the total

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cumulative hours. Testing shall continue until the total unit hours together with the total count of relevant equipment failures permit either an accept or reject decision in accordance with Figure 5. Only equipment "on" time shall contribute to the total unit hours. No single equipment's "on" time shall contribute to the total unit hours. No single equipment's "on" time shall be less than one-half the average operating time of all equipment on test.

4.8 Preconditioning. The scanner shall be mounted on a suitable fixture and connected to a scan-interlace module or similar circuit. Power shall be applied and the scanner shall be operated for at least 8 hours. Monitoring is not required during preconditioning (see 3.12).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's system Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory in the performance of the contract.)

6.1 Intended use. The Scanner is intended for infrared systems. When the scanner is mounted in an infrared (IR) system, the front side of the scan mirror directs incoming IR energy through a set of IR imaging optics onto an array of IR detectors. The back side of the mirror directs the visual output from a light emitting diode (LED) array through a set of visible collimating optics and into a visible display optics assembly.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of supplemental specification sheet.

c. First article tests are required.

1) Time frame for submission of first article test reports when first article tests are required.

2) Time frame for approval of first article test reports when first article tests are required.

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d. Production delivery schedule.

1) Defined in terms of monthly lots.

2) Actions required in case of Group C failures.

e. Preservation and packaging procedures (see Section 5).

g. If Scanners are procured as part of end item assemblies, temperature shock, shock and vibration may not be required unless otherwise specified.

h. When ordering a specific type of scanner, only those parameters in each test group that are applicable to that scanner may be required and may be specified in a supplemental specification.

i. Scan-Interlace PL-1403/UA and PL-1403A/UA per SM-D-773950, 5008929, respectively, and/or Scan-Interlace PL-1408/UA per 5008646 may be ordered from any existing supplier of these modules.

j. Deliver all first article samples to the contracting officer.

k. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods .

l. Group C inspection monthly sample size to be established based on cumulative totals of concurrent contracts.

m. The preconditioning requirement of 3.12 is required for procurement of all modules to this specification made directly for the Government. Implementation of preconditioning may be waived by the Government. The preconditioning requirement may be waived by system contractors ordering modules to this specification for use in the manufacture of complete systems if these systems or subassemblies thereof are subjected to equivalent preconditioning.

6.3 Subject term (key word) listing.

Scan mechanism
Scan-Interlace

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR

(Project 5855-0220)

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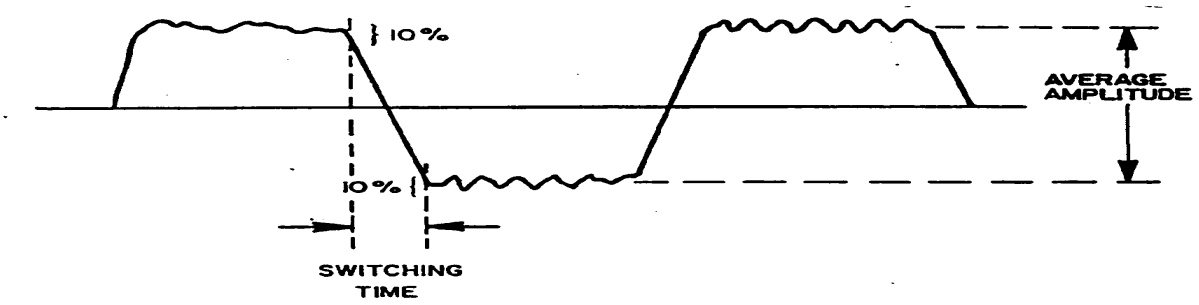


FIGURE 1. INTERLACE TRANSDUCER OUTPUT.

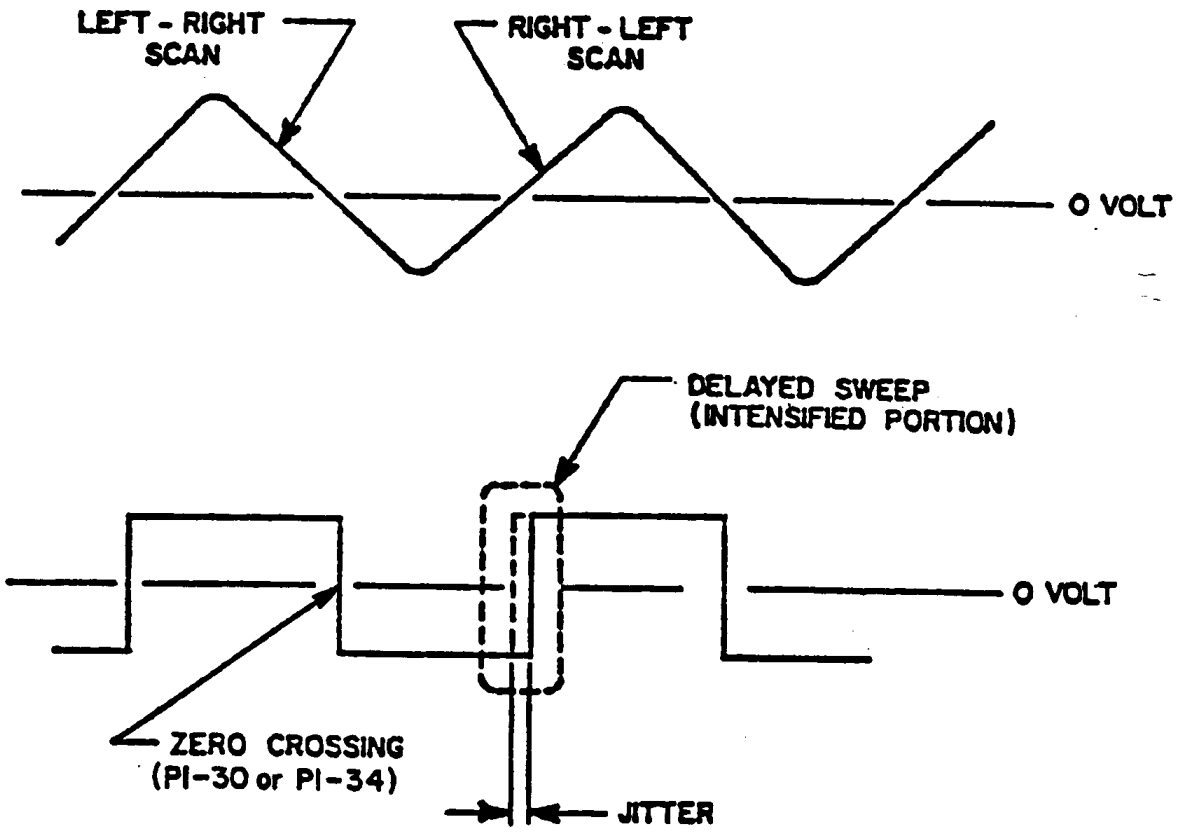


FIGURE 2. ZERO CROSSING WAVEFORM JITTER.

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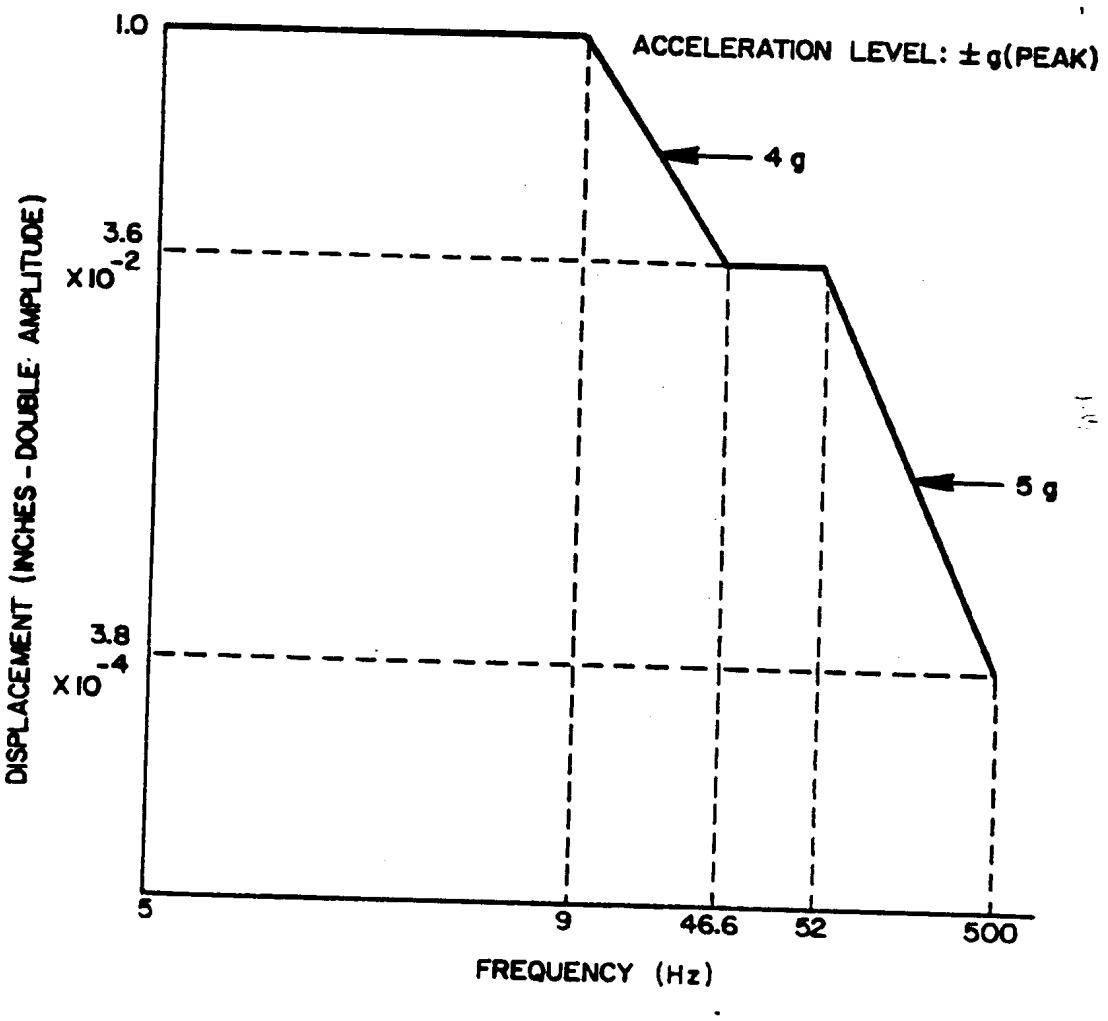


FIGURE 3. VIBRATION TEST PROFILE .

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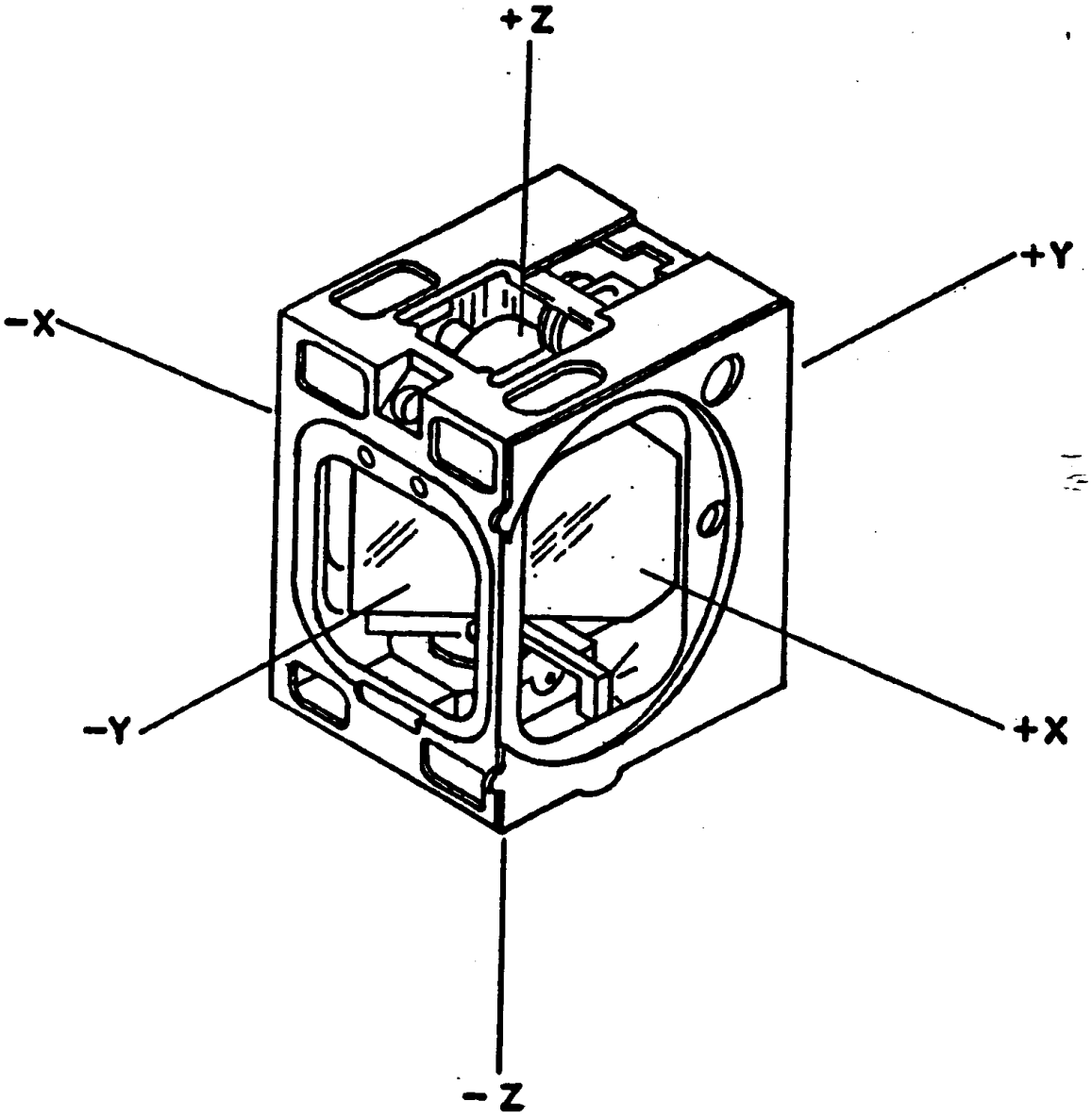
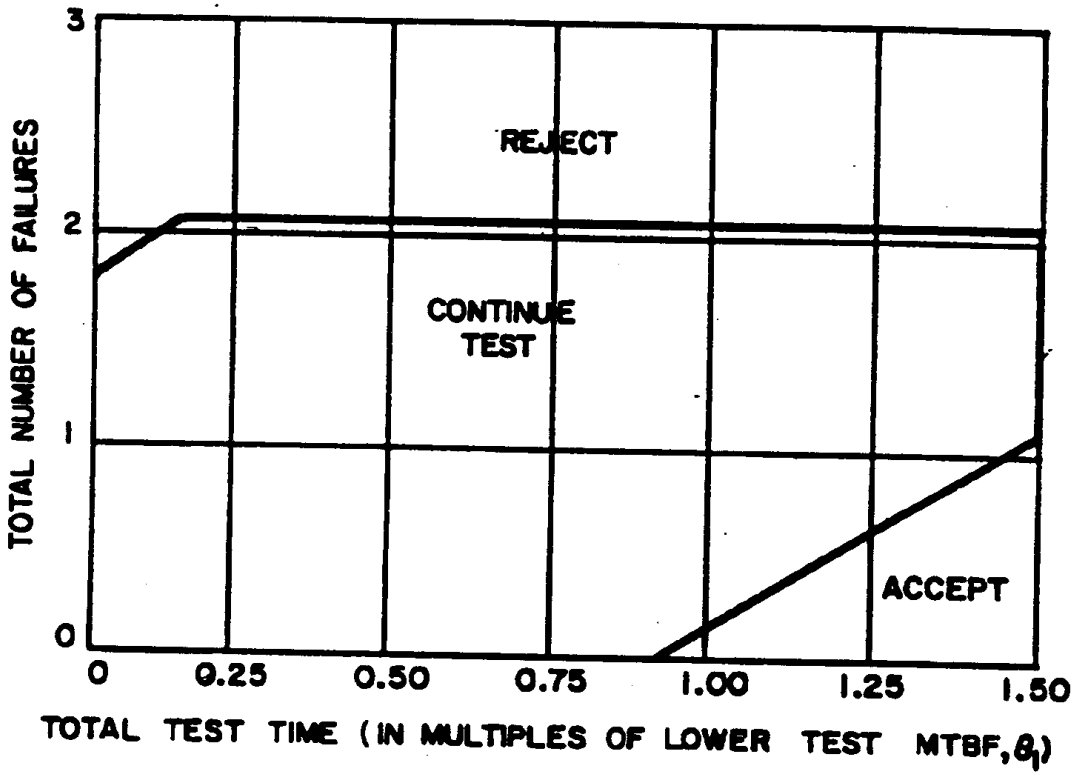


FIGURE 4. SCANNER GEOMETRIC REFERENCES.

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NUMBER OF FAILURES	TOTAL TEST TIME* REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	NA	0.89
1	NA	1.44
2	0.12	1.50
3	1.50	NA

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF EQUIPMENT "ON TIME" AND IS EXPRESSED IN MULTIPLES OF LOWER TEST MTBF, θ_1

FIGURE 5. FIRST ARTICLE RELIABILITY, ACCEPT-REJECT CRITERIA.

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Appendix A
PROCEDURE FOR VIBRATION VERIFICATION

A.1 SCOPE

A1.1 Scope. This appendix details the procedure for vibration verification. This appendix is a mandatory part of this specification.

1. PURPOSE. The vibration test is performed to determine if equipment is constructed to withstand expected dynamic vibrational stresses and to insure that performance degradations or malfunctions will not be produced by the service vibration environment. Tests specified herein are established for equipment which may be used in a variety of military applications.

2. APPARATUS. Vibration equipment with required instrumentation.

3. GENERAL. The vibration test tables and figures provide a convenient means of summarizing test procedures to be specified in the equipment specification or test plan according to various military applications. Each table title refers to the applicable category of the equipment to be tested. Guidance for selecting a test is as follows:

a. Determine equipment category (or categories) in accordance with 3.1.

b. Proceed to the designated table(s) and figure(s) corresponding to the applicable equipment category. Select a test based on the table instructions, procedure, and figure designated.

3.1 Equipment category. For purposes of this test method, equipment is categorized according to the vehicle in which it will be installed or transported as follows:

Category	Description	Procedure	Table	Figure
a	Equipment installed in airplanes and helicopters. An equipment intended for both aircraft and helicopters should be qualified for each installation in turn.	--	514.1-I (discontinued)	514.1-1 (discontinued)
b.1	Equipment installed in propellor airplanes and equipment mounted directly to engines including jet engines.	I	514.2-II	514.2-2
b.2	Equipment installed in jet airplanes except for jet engine mounted equipment.	IA	514.2-IIA	514.2-2A

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c	Equipment installed in helicopters	I	514.2-III	514.2-3
d.1	Equipment installed in external stores carried on airplanes	IIA	514.2-IV	514.2-4
d.2	Assembled external stores carried on airplanes	IIB	514.2-IV	514.2-4A
d.3	Assembled externally carried stores for helicopters	IIC	514.2-IVA	514.2-4B, 4C, 4D, 4E, 4F
e	Equipment installed in ground launched missiles	V, VI or VII as applicable	514.2-V	514.2-5
f	Equipment installed in ground vehicles	VIII	514.2-VI	514.2-6
g	Equipment transported as cargo	X, XI	514.2-VII	514.2-7
h	Ground equipment, excluding category f. (For transportation see category g.)	--	--	--
i	Shipboard and amphibious equipment or when a ship is the common carrier (see 4.6.13)	--	--	--

3.2 Applicable tests. For any given equipment category, all tests listed beside the selected procedure for the applicable equipment mounting configuration in the tables shall be performed unless otherwise specified. For example, referring to table 514.2-V section A, for testing equipment category e when procedure VI is selected, there are three parts with different test levels indicated by the test curves. Tests indicated by (X) in all three parts shall be performed to evaluate equipment with isolators.

3.3 Selection of test curves. A curve shall be selected from the tables and figures or by making a detailed analysis of the expected vibration environment within the particular vehicle involved. A primary consideration is the equipment location with respect to predominant vibration sources such as high intensity noise of jet and rocket exhausts, aerodynamic excitation including atmospheric wind and turbulence, and unbalance of rotating

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parts. Additional factors to be considered shall include attenuation or amplification and filtering by structural members. Guidance for selecting vibration curves with respect to equipment location or application is given in the tables. Applicable test curves for each equipment category are shown on the figures.

3.4 Procedure selection. The equipment specification or test plan shall identify which tests are to be imposed on the equipment by specifying the applicable procedure, table, figure, and test curve.

3.4.1 Example. Select the test conditions for equipment to be used in the following application:

Category: Equipment installed in a propellor airplane

Equipment location: Forward half of fuselage

Equipment mounting: On vibration isolated panel

Referring to table 514.2-II, the above identification specifies the following test conditions:

Procedure I

Part 1 (curve C)

Part 2 (curve B)

Part 1 specifies a resonance search, resonance dwell and sinusoidal vibration cycling to the level of curve C from figure 514.2-2 within the time schedule specified for part 1 on table 514.2-II. Next, with vibration isolators removed in accordance with note 2, part 2 is performed the same as part 1 but to the test level or curve B from figure 514.2-2 within the time schedule specified for part 2 from table 514.2-II.

4. PROCEDURES. The basis for selecting a test procedure for a particular equipment category shall be according to 3. The vibration environment, specified by the curve selected from applicable tables in accordance with 3, shall be applied to each of the three mutually perpendicular axes of the test item. (For assembled external stores, see procedure IIB.) The entire sequence of tests may be accomplished for any one axis before changing to the next axis. The transverse motion at the input monitoring point(s) shall be minimized, and should be limited to 100 percent of the input motion except that reaction machines shall be balanced to reduce transverse motion ±10 percent.

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4.1 Test item operation. Unless otherwise specified, the test item shall be operated during application of vibration (resonance search, resonance dwell, cycling, and random vibration) so that functional effects caused by these tests may be evaluated. Procedure IA, IIA and IIB provide for a functional vibration test and an endurance vibration test. The test item shall meet performance requirements as specified in the detail equipment specification and General Requirements, 3.2 while the functional vibration test levels are being applied. When a test item performance test is required during vibration and the time required for the performance test is greater than the duration of the vibration test, the performance test may be abbreviated accordingly. The test item shall be operated and inspected and the results shall be obtained in accordance with General Requirements, 3.2

4.2 Mounting techniques. The test item shall be attached by its normal mounting means, either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of extraneous responses in the test setup. The test load shall be distributed as uniformly as possible on the vibration exciter table in order to minimize effects of unbalanced loads. The input control sensing device(s) shall be rigidly attached to the vibration table, or fixture if used, as near as possible to the attachment point(s) of the test item. Additional vibration sensors shall be located in or on the test item to determine resonant frequencies and amplification factors. Locations to be selected should include main structure, printed circuit boards, large components, and modules, where practicable. The sensor sizes and weights shall be limited so that their effect on the dynamic responses being measured is minimal. For sinusoidal vibration, when necessary for obtaining uniform results, a tracking filter should be used in the vibration exciter control feedback loop prior to the servo input.

4.3 Combined temperature-vibration test. Tests shall be performed under room ambient conditions unless a high or low temperature vibration test is specified, in which case the temperature extremes and time duration also shall be as specified in the equipment specification.

4.5 Common test techniques

4.5.1 Sinusoidal vibration tests. The vibration shall be applied along each of three mutually perpendicular axes of the test item. The vibratory acceleration levels or double amplitudes of the specified test curve shall be maintained at the test item mounting points. When necessary for obtaining uniform results, a tracking filter should be used in the exciter control feedback loop prior to the servo input. When specified, for

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sinusoidal resonance search, resonance dwell, and cycling tests of items weighing more than 80 pounds mounted in airplanes, and missiles, the vibratory accelerations shall be reduced 1g for each 20-pound increment over 80 pounds. Acceleration derating shall apply only to the highest test level of the selected curve, but in no case shall the derated test level be less than 50 percent of the selected curve (see note 1 of the applicable table). When the input vibration is measured at more than one control point, the control signal shall be the average of all the accelerometers unless otherwise specified. For massive test items, fixtures and large force exciters, it is recommended that the input control level be an average of at least three or more inputs.

4.5.1.1 Resonance search. Resonant frequencies of the equipment shall be determined by varying the frequency of applied vibration slowly through the specified range at reduced test levels but with sufficient amplitude to excite the item. Sinusoidal resonance search may be performed using the test level and cycling time specified for sinusoidal cycling test, provided the resonance search time is included in the required cycling test time of 4.5.1.3.

4.5.1.2 Resonance dwell. The test item shall be vibrated along each axis at the most severe resonant frequencies determined in 4.5.1.1. Test levels, frequency ranges, and test times shall be in accordance with the applicable conditions from the tables and figures for each equipment category. If more than four significant resonant frequencies are found for any one axis, the four most severe resonant frequencies shall be chosen for the dwell test. If a change in the resonant frequency occurs during the test, its time of occurrence shall be recorded and immediately the frequency shall be adjusted to maintain the peak resonance condition. The final resonant frequency shall be recorded.

4.5.1.3 Cycling. The test item shall be vibrated along each axis in accordance with the applicable test levels, frequency range, and times from the applicable tables and figures. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 514.2-10. The specified sweep time is that of an ascending plus a descending sweep and is twice the ascending sweep time shown on figure 514.2-10 for the specified range.

4.5.2 Random vibration test. The test item shall be subjected to random vibration along each of three mutually perpendicular axes according to the specified curve. Test times shall be according to the applicable schedule from the tables. The instantaneous random vibration acceleration peaks may be limited to three times the rms acceleration level. The power spectral density of the test control signal shall not deviate from the specified requirements by more than +100,-30 percent (+3, -1.5 dB) below 500 Hz

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and +100, -50 percent (±3 dB) between 500 Hz and 2,000 Hz except that deviations as large as +300, -75 percent (±6 dB) shall be allowed over a cumulative bandwidth of 100 Hz maximum, between 500 and 2,000 Hz.

Tolerance levels in terms of dB are defined as:

$$dB = 10 \log_{10} \frac{W_1}{W_0}$$

Where W_1 = measured acceleration power spectral density in G^2/Hz units. The term W_0 defines the specified level in G^2/Hz units. Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant product, BT = 50, minimum. Specific analyzer characteristics shall be as specified below or equivalent, subject to the BT = 50, minimum limitation.

a. On-line, contiguous filter, equalization/analysis system having a bandwidth as follows:

B = 25 Hz, maximum between 20 and 200 Hz

B = 50 Hz, maximum between 200 and 1,000 Hz

B = 100 Hz, maximum between 1,000 and 2,000 Hz

b. Swept frequency analysis systems characterized as follows:

(1) Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B = 25 Hz, maximum between 20 to 200 Hz

B = 50 Hz, maximum between 200 to 1,000 Hz

B = 100 Hz, maximum between 1,000 to 2,000 Hz

(b) Analyzer averaging time = T = 2 RC = 1 second, minimum, where T = True averaging time and RC = analyzer time constant

(c) Analysis sweep rate (linear) = $R = \frac{B}{4RC}$ or $\frac{B^2}{8}$ (Hz/second) maximum, whichever is smaller.

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(2) Constant percentage bandwidth analyzer

(a) Filter bandwidth = pf_c = one-tenth of center frequency maximum, ($0.1f_c$) where p = percentage and f_c = analyzer center frequency

(b) Analyzer averaging time = $T = \frac{50}{pf_c}$, minimum

(c) Analysis sweep rate (logarithmic) = $R = \frac{pf_c}{4RC}$ or $\frac{(pf_c)^2}{8}$

(Hz/second), maximum, whichever is smaller.

c. Digital power spectral density analysis system employing quantization techniques providing accuracies corresponding to the above approach.

Accelerometer(s) employed for test level control shall be mounted in accordance with 4.2. Where more than one accelerometer is employed for test level control, the power average of the several accelerometer signals shall be used as the test level signal control.

4.6 Procedures

4.6.1 Procedure I. Equipment installed in airplanes or helicopters. (Category b.1 or c.)

4.6.1.1 Part 1. For category b.1 equipment, proceed the same as in 4.5.1.1, 4.5.1.2, and 4.5.1.3. The test levels and time durations shall be as specified in table 514.2-II and figure 514.2-2. For category c equipment, proceed the same as in 4.5.1.3 only. The test levels and time schedules shall be as specified in table 514.2-III and figure 514.2-3.

4.6.1.2 Part 2. For category b.1 equipment, test items normally provided with vibration isolators shall be vibrated in accordance with 4.5.1.1, 4.5.1.2, and 4.5.1.3 with the vibration isolators removed but including any other required holding devices. The test levels and time durations shall be as specified in table 514.2-II and figure 514.2-2. For category c equipment, proceed the same as in 4.5.1.3 only with the vibration isolators removed but including any other holding devices. The test levels and time schedules shall be as specified in table 514.2-III and figure 514.2-3.

4.6.2 Procedure IA. Random vibration test for equipment installed in jet airplanes. (Not for turboprop aircraft or jet powered helicopters.) The random vibration environment which occurs at equipment locations in jet aircraft stems from four principal sources:

a. Turbulent aerodynamic airflow along external surfaces of the aircraft structure.

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b. Jet engine noise impinging on aircraft structure.

c. Gun blast pressure impinging on aircraft structure from high speed repetitive firing of installed guns.

d. General aircraft motions caused by such factors as runway roughness, landing, and gusts.

The tests outlined in this procedure consider all of these environments and require design to the most severe of these. These tests are preferred for use with equipment in jet aircraft in lieu of the sinusoidal tests of procedure I, table 514.2-II, figure 514.2-2, except for jet engine mounted equipment. For equipment mounted directly to aircraft jet engines, use procedure I. To determine an equipment specific random vibration test, compute functional and endurance test levels for aerodynamic induced and for jet engine induced vibration from table 514.2-IIA and figure 514.2-2A. Use the more severe of the two functional levels as the equipment's functional test, and the more severe of the two endurance levels (on an equal time, T, basis) for the equipment's endurance test.

4.6.2.1 Performance of test. The individual equipment test item shall be subjected to broadband random vibration excitation. The power spectral density tolerances of applied vibration shall be according to 4.5.2. The test item shall be attached to the vibration exciter according to 4.2. Equipment hard mounted in service shall be hard mounted to the test fixture. Equipment isolated in service shall use service isolators when mounted on the test fixture. If service isolators cannot be made available during the qualification test, isolators shall be provided with characteristics such that the isolator/equipment resonant frequencies shall be between 20 Hz and 45 Hz with resonant amplification ratio between 3 and 5. Vibration shall be applied sequentially along each of the three orthogonal axes of the test item. Two test levels are required, a functional level and an endurance level. For each axis, one half of the functional test shall be conducted first, then the endurance test, followed by the second half of the functional test. The equipment shall perform according to the equipment specification operating requirements in accordance with General Requirements, 3.2 during the functional testing. The acceleration power spectral density (G^2/Hz) of applied vibration, as measured on the test fixture at mounting points of the test item, shall be according to table 514.2-IIA and figure 514.2-2A. The functional and endurance test time durations and other test conditions shall be determined from the test level equations and other parameter values from table 514.2-IIA.

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4.6.2.2 Equipment with isolators. Equipment designed for operational installation on vibration isolators shall also be subjected to a minimum rigidity endurance test with the isolators removed. This test shall be conducted according to part 2, procedure 1, table 514.2-II, and curve AR of figure 514.2-2. At the conclusion of this test the equipment shall provide specified performance, in accordance with General Requirements, 3.2.

4.6.2.3 References

a. Dreher, J.F., "Aircraft Equipment Random Vibration Test Criteria Based on Vibrations Induced by Turbulent Airflow Across Aircraft External Surfaces," Shock and Vibration Bulletin No. 43 (part 3, pp 127-139), NRL, Wash., DC, June 1973.

b. Wafford, J.H., and Dreher, J.F., "Aircraft Equipment Random Vibration Test Criteria Based on Vibration Induced by Jet and Fan Engine Exhaust Noise," Shock and Vibration Bulletin No. 43 (part 3, pp 141-151), NRL, Wash., DC, June 1973.

c. Earls, D.L., "Technical Progress on New Vibration and Acoustic Tests," for proposed MIL-STD-810C entitled, Environmental Test Methods, Journal of Environmental Sciences, pp 22-32, July/August 1973.

4.6.3 Procedure II. This procedure includes tests for three applications:

a. Individual equipment items designed for installation in external stores carried on airplanes

b. Assembled external stores carried on airplanes

c. Assembled external stores carried on helicopters

These tests apply to such stores as air launched missiles, bombs, dispensers, instrument pods and rocket launchers (see equipment categories d.1, d.2 and d.3 as applicable).

4.6.3.1 Procedure IIA. Equipment installed in external stores carried on airplanes.

4.6.3.1.1 Test conditions. The individual equipment test item (designed for installation in an external store) shall be subjected to broadband random vibration excitation. The power spectral density tolerances of applied vibration shall be according to 4.5.2. The test item shall be attached to the vibration exciter according to 4.2. Vibration shall be applied sequentially along each of the three orthogonal axes of the test item.

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4.6.3.1.2 Captive flight. Two test levels are required, a functional level and an endurance level. For each axis, one-half of the functional test shall be conducted first, then the endurance test, followed by the second-half of the functional test. The equipment shall perform according to the equipment specification operating requirements in accordance with General Requirements, 3.2 during the functional testing. The acceleration power spectral density (G^2/Hz) of applied vibration, as measured on the test fixture at mounting points of the test item, shall be according to table 514.2-IV and figure 514.2-4 except as noted below. The functional and endurance test time durations and other test conditions shall be determined from the test level requirements and other parameter values from table 514.2-IV. If the computed functional and endurance (T=1) test levels (W_2) are less than 0.04 G^2/Hz, use W_2 = 0.04 G^2/Hz and T = 1 for the endurance test.

4.6.3.1.3 Free flight functional test. For stores that are deployed by separation from the aircraft (free flight) such as bombs and missiles, a free flight functional test shall be conducted in addition to the captive flight tests of 4.6.3.1.2. The equipment shall perform according to the equipment specification operating requirements in accordance with General Requirements, 3.2 during the functional testing. 4.6.3.1.1, 4.6.3.1.2, table 514.2-IV, and figure 514.2-4 shall be used to determine the test procedures, levels and frequency spectra for the free flight test except as noted below. In this case, factors A_1, A_2, and (N/3T) shall be set equal to one. The value of q shall be the maximum value attainable during free flight. The duration of this functional test, per axis, shall equal the maximum free flight time expected at maximum q, but not less than 30 seconds. In the event that all free flight functional checks are made during the captive functional test and the captive functional test levels are larger than or equal to those derived here (4.6.3.1.3), no free flight functional test is required.

NOTE: Items marked DISCONTINUED refer to items of the superseded issue of MIL-STD-810 including notices thereto.

4.6.3.1.4 References

a. Dreher, J. F., Lakin, E. D., Tolle, E. A., "Vibracoustic Environment and Test Criteria for Aircraft Stores During Captive Flight," Shock and Vibration Bulletin No. 39, Supplement, (pp 15-40), NRL, Wash., DC, April 1969.

b. Dreher, J. F., "Effects of Vibration and Acoustical Noise on Aircraft/Stores Compatibility," Proc. Aircraft Store Compatibility Symp., Vol. 6, pp 245-272, Eglin AFB, Florida, November 1969.

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c. Piersol, A. G., "Vibration and Acoustic Test Criteria for Captive Flight of Externally Carried Aircraft Stores," AFFDL-TR-71-158, Wright-Patterson AFB, Ohio, December 1971.

d. Burkhard, A., "Captive Flight Acoustic Test Criteria for Aircraft Stores," Shock and Vibration Bulletin No. 43, Part 3 (pp 113-126), NRL, Wash., DC., June 1973.

4.6.3.2 Procedure IIB. Assembled externally carried airplane stores

4.6.3.2.1 Purpose. This vibration test is performed to determine that the assembled store as a system is constructed to withstand and perform in the expected dynamic environment. The nature of this test is to simulate the lower frequency vibratory environment of the store by application of vibratory energy at discreet points. To insure that the vibratory test levels reflect the inflight vibratory levels, the selection of the test points and vibration input levels are determined by the response of the store. Procedure II of Method 515.2, acoustic testing for assembled externally carried aircraft stores, shall also be conducted to simulate the higher frequency distributed aerodynamically induced vibratory environment. Acoustic testing is not required if the minimum value of f_o of 4.6.3.2.3 is greater than 1,200 Hz. Procedure IIA of this method shall be used when individual equipment items such as fuzes, electronic equipment, etc., need to be tested separately from the whole store.

4.6.3.2.2 Accelerometer placement. Accelerometers to monitor the vibratory response of the store shall be mounted on two relatively hard points or rings within the store: one in the nose section and one in the aft section. For stores such as bombs with nonintegral tail cones, the aft section mounting point shall be in the aft most section of the main body of the store. At each mounting point or ring, two accelerometers shall be mounted - one in the vertical and one in the lateral plane. (Longitudinal direction is along the axis of the store. The vertical direction shall be considered as perpendicular to the longitudinal axis and contained in a plane passing through the mounting lugs.)

4.6.3.2.3 Test levels. The test frequency spectrum for each store mounted monitoring accelerometer shall be determined from figure 514.2-4A and the values of table 514.2-IV. Generally, the test spectrum for the forward accelerometers will be different from that of the aft accelerometers. Test levels for both a functional and endurance test shall be determined. The value of f_o used in figure 514.2-4A for this test shall be defined as follows:

$f_o = (t/R^2) \times 10^5 + 100$ Hz for stores with circular or elliptical cross-sections

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f_o = 500 Hz for all other cross-sections

t = local store average skin thickness where radius, R, is measured--inches

R = one-half the average of major and minor external diameters (inches) of the elliptical cross-section (for a cylindrical section use local average radius; for conical section use smallest f_o calculated using geometry within 1 foot of accelerometer mounting point)

NOTE: If calculated $f_o \leq 1,200$ Hz, use 2,000 Hz
Acceptable range for $t/R^2 = .001 \leq t/R2 \leq .02$
If calculated values fall outside these limits, use these limit values.

4.6.3.2.4 Test setup. The store shall be mounted and tested with its longitudinal axis horizontal and positioned so that its mounting lugs are on top of the store using 4.6.3.2.4a or, as an alternate, 4.6.3.2.4b, depending on the facilities available.

a. Mounting method. The store shall be suspended from a structural frame by means of its normal mounting lugs using hooks and sway braces which simulate the operational mounting apparatus. The test setup shall be such that the rigid body modes (translation and rotation) of vibration for the store/frame/suspension system are between 5-20 Hz. Vibration shall be applied to the store by means of a rod or other suitable mounting device running from a vibration shaker to a relatively hard structurally supported point on the surface of the store.

b. Alternate mounting method. The store shall be hard mounted directly to the shaker, using its normal mounting lugs and a suitable fixture. The stiffness of the mounting fixture shall be such that its induced resonant frequencies are as high as possible but none are below one-third of any f_o frequency.

4.6.3.2.5 Determination of resonance response peaks. Vibration shall be applied at an input level sufficient to excite vibratory responses within the store. The frequencies at which significant resonance response peaks occur shall be identified for each store mounted accelerometer using no greater than a 10 percent constant percentage bandwidth analysis. A resonance response peak shall have at least a 6 dB amplification of input level to be considered significant. This shall be done for both vertically and laterally applied vibration.

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4.6.3.2.6 Captive flight test. The following procedure shall be performed so that both the lateral and vertical directions are tested to their respective test levels. Separate testing for each direction shall be required unless one test is able to excite both directions simultaneously to their respective test levels. Two test levels are required, a functional level and an endurance level. For each axis, one-half of the functional test shall be conducted first then the endurance test followed by second half of functional test. The equipment shall perform according to the equipment specification operating requirements during functional testing.

4.6.3.2.6.1 Part 1. Random vibration shall be applied to the store using an input spectrum shape of the store mounted forward accelerometer, but at an input level 6 dB down from the calculated response level of the forward accelerometer. The input random vibration level shall then be peaked or notched in those portions of the frequency range where significant resonance response peaks were identified in 4.6.3.2.5. The peaking and notching of the input spectrum shall be such that the significant resonance response peaks of each store mounted accelerometer in the direction of applied vibration shall be equal to or greater than their respective test levels calculated in 4.6.3.2.3 and figure 514.2-4A. When using the test setup of 4.6.3.2.4a, different attachment locations for the shaker to the store may be tried to determine an optimum point so that both ends of the store are simultaneously tested to their respective test levels.

4.6.3.2.6.2 Part 2. The response of the accelerometers in the direction perpendicular to the store mounted accelerometers which were used to establish the test spectrum shall be compared to their levels specified in 4.6.3.2.3. For each frequency where the response of a (the) perpendicular accelerometer(s) is above their specified levels the following procedure may be used.

4.6.3.2.6.3 Part 3. Calculate at each of these frequencies a ratio of specified to observed levels for each accelerometer which was in the direction of vibration and those perpendicular accelerometer(s) which have excessive levels. These ratios shall be averaged so that an average value is determined for each frequency. The input vibration spectrum shall be adjusted so that at each of these frequencies their respective average value shall be equal to unity. The duration of the functional and endurance test is given in table 514.2-IV and the test levels for each test are determined in 4.6.3.2.3.

4.6.3.2.7 Free flight functional test. For stores that are deployed by separation from the aircraft (free flight) such as bombs and missiles, a free flight functional test shall be conducted in addition to the captive flight tests of 4.6.3.2.6. The equipment shall perform according to the equipment specification operating requirements as specified in General Requirements, 3.2 during the functional testing. Use 4.6.3.2.6, table 514.2-IV, and figure 514.2-4A to determine the test procedures, levels and frequency spectra for the free flight test except as noted below.

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In this case, factors A_1, A_2, and (N/3T) shall be set equal to one. The value of q shall be the maximum value attainable during free flight. The duration of this functional test, per axis, shall equal the maximum free flight time expected at maximum vibration level, but not less than 30 seconds. In the event that all free flight functional checks are made during the captive functional test and the captive functional test levels are larger than or equal to those derived here (4.6.3.2.7), no free flight functional test is required.

4.6.3.2.8 References

a. Dreher, J. F.; Lakin, E. D.; Tolle, E. A.; "Vibracoustic Environment and Test Criteria for Aircraft Stores During Captive Flight," Shock and Vibration Bulletin No. 39, Supplement (pp 15-40), NRL, Wash., DC, April 1969.

b. Dreher, J. F., "Effects of Vibration and Acoustical Noise on Aircraft/ Stores Compatibility," Proc. Aircraft Store Compatibility Symp., Vol. 6, pp 245-272, Eglin AFB, Florida, November 1969.

c. Piersol, A. G., "Vibration and Acoustic Test Criteria for Captive Flight of Externally Carried Aircraft Stores," AFFDL-TR-71-158 Wright-Patterson AFB, Ohio, December 1971.

4.6.3.3 Procedure IIC, assembled externally carried stores for helicopters

4.6.3.3.1 Purpose. To determine that externally carried stores attached to the wings and pylons of helicopters (with a main rotor speed of 324 rpm, two blades, such as UH-1B, UH-1C, AH-1G, etc) are capable of withstanding expected dynamic vibrational stresses normally induced by the helicopter during flight conditions and to qualify the equipment for service life. If the equipment normally operates while being carried by the helicopter, then the equipment is expected to operate satisfactorily during the vibration test. For helicopters other than the above, use values of frequencies and g levels applicable to the helicopter on which that item is to be mounted.

NOTE: This procedure is only applicable to the operating phase of the equipment and does not include the common carrier transportation phase.

4.6.3.3.2 Test fixture. A rigid test fixture shall be designed for attaching the test item to the vibration exciter in all three axes with the mounting lugs in the up position. The test item shall be attached to the fixture by its normal mounting means (e.g., suspension lugs for 2.75-inch FFAR launchers). The vibration levels shall be controlled from an accelerometer mounted at or near the attachment point on the test item.

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4.6.3.3.3 Cycling test. The test item shall be subjected to vibration cycling in each of three orthogonal axes. The frequency of applied vibration shall be cycled at a logarithmic rate between the frequency limits and at the vibratory acceleration levels as shown on figure 514.2-4B. The cycling period and rate shall be determined from table 514.2-IVA.

4.6.3.3.4 Dwell test. The test item shall be subjected to vibration dwell in each of three orthogonal axes. The dwell test shall be conducted at the four frequencies and at the vibratory acceleration levels as shown on figures 514.2-4D, 514.2-4E, 514.2-4F for the particular weight of the test item. (The minimum acceleration level shall be 0.5g for all dwell tests.) The dwell period at each frequency shall be determined from curve A of figure 514.2-4C.

NOTE: For test items such as rocket launchers, which are flown to the target area loaded and come back empty, it is recommended that one-half of the cycling test be applied to a loaded launcher and the other half be applied to an empty launcher.

4.6.4 Procedure III (Former 4.8 DISCONTINUED). (See equipment categories d.1 and d.2.)

4.6.5 Procedure IV (Former 4.9 DISCONTINUED). (See equipment categories d.1 and d.2.)

4.6.6 Procedure V. Equipment installed in ground launched missiles (without vibration isolators, equipment category e.)

4.6.6.1 Part 1. Proceed the same as in 4.5.1.3. The test level shall be according to one specified curve P through U from table 514.2-V and figure 514.2-5. Test time schedules shall be as listed for part 1 of procedure V as specified in table 514.2-V.

4.6.6.2 Part 2. Proceed the same as in 4.5.2. The test level shall be according to one specified in curve AE through AP from table 514.2-V and figure 514.2-5. Test time schedules shall be as listed for part 2 of procedure V as specified in table 514.2-V.

4.6.7 Procedure VI. Equipment installed in ground launched missiles (with vibration isolators, equipment category e.)

4.6.7.1 Part 1. Test items normally provided with vibration isolators shall be vibrated with the isolators in place as in 4.5.1.3. Test levels shall be according to one specified curve P through U from figure 514.2-5. Test time schedules shall be as listed for part 1 of procedure VI as specified in table 514.2-V.

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4.6.7.2 Part 2. Test items normally provided with vibration isolators shall be vibrated in accordance with 4.5.1.3 with the vibration isolators removed but including any other required holding devices. Test levels shall be according to curve N from figure 514.2-5. Test time schedules shall be as listed for part 2 of procedure VI as specified in table 514.2-V.

4.6.7.3 Part 3. With vibration isolators in place, proceed the same as in 4.5.2. Test levels shall be according to one specified curve AE through AP from figure 514.2-5. Test time schedules shall be as listed for part 3 of procedure VI as specified in table 514.2-V.

4.6.8 Procedure VII (Former 4.12). Equipment installed in ground launched missiles. (Isolated equipment, tested without isolators, equipment category e.)

4.6.8.1 Part 1. Proceed the same as in 4.5.1.3. Test levels shall be according to curve N from figure 514.2-5. Test time schedules shall be as listed for part 1 of procedure VII as specified in table 514.2-V.

4.6.8.2 Part 2. Proceed the same as in 4.5.2. Test levels shall be according to curve AE from figure 514.2-5. Test time schedules shall be as specified for part 2 of procedure VII as specified in table 514.2-V.

4.6.9 Procedure VIII (Former 4.13). Equipment installed in ground vehicles (equipment category f). Proceed the same as in 4.5.1.3. Test levels shall be according to one specified curve V, W, or Y from figure 514.2-6. Test time schedules shall be as specified in procedure VIII as specified in table 514.2-VI. When test item resonances below 5 Hz are measured or expected the test curves shall be extended to 2 Hz and the sweep time shall be 18 minutes (2-500-2 Hz); 15 minutes (2-200-2 Hz).

4.6.10 Procedure IX (Former 4.14)

4.6.10.1 Part 1 (Former 4.14.1) DISCONTINUED

4.6.10.2 Part 2, bounce, vehicular

4.6.10.2.1 Apparatus. A package tester capable of 1 inch (double amplitude) displacement and of suitable capacity for testing military equipment.

4.6.10.2.2 Test conditions

a. Cover the test bed of the package tester with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Secure the plywood with six-penny nails, with top of heads flush with, or slightly below the surface. Space nails at 6-inch intervals around all four edges. If the distance between either pair of fences is greater than 24 inches, the plywood shall also be nailed at 3-inch intervals in a 6-inch square at the center of the test area.

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b. Using suitable wooden fences, constrain the vehicular, or simulated, adapter plate to a horizontal motion of not more than 2 inches in any lateral direction.

4.6.10.2.3 Performance of test

Step 1 - Prepare the test item in accordance with General Requirements, 3.2, securing the test item to the vehicular, or simulated, adapter plate and placing on the package tester with the constraints outlined in 4.6.10.2.2. If the test item weighs over 200 pounds, simulated adapter plate approved by the procuring agency shall be used.

Step 2 - Attach an accelerometer as close as possible to the point of test item attachment to record the shock transmitted to the test item.

Step 3 - Adjust the package tester, shafts in phase and table operating in a vertical linear mode, to a speed such that the average value of the random acceleration peaks shall be 7.5 ±2.5g's. Measure this input with an accurate measuring or recording system at the output of a band pass filter. The filter band pass shall be 0.2 to 100 Hz and the attenuation slope shall be 12 to 18 dB per octave at the 3-dB down point. Due to the random nature of the input, pulses greater than 10g's can be expected to occur, however, if they are infrequent, they need not be used in calculating the average. Perform the test for a total of 3 hours. At the end of each 3/4-hour period, rotate the adapter plate and test item 90 degrees each time in the same direction.

Step 4 - At the end of the 3-hour period, operate and inspect the test item in accordance with General Requirements, 3.2. Then inspect the test item as specified in General Requirements, 3.2.

4.6.11 Procedure X (Former 4.15). Equipment transported as secured cargo. Proceed the same as in 4.5.1.3 with the unpackaged test item secured to the vibrator table. The test level and time schedules shall be in accordance with figure 514.2-7 and table 514.2-VII. The test item will not be operated during vibration.

4.6.12 Procedure XI (Former 4.16). Equipment transported as loose cargo.

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4.6.12.1 Part 1 (Former 4.16.1) DISCONTINUED

4.6.12.2 Part 2, bounce, loose cargo

4.6.12.2.1 Purpose. To determine that the equipment, as prepared for field use, shall be capable of withstanding the vibrations normally induced during combat transportation as loose cargo. Equipment in this class is normally transported in a transit case, combination case, or special container from which it is removed just prior to use, e.g., ammunition or missiles.

4.6.12.2.2 Apparatus. A package tester capable of 1 inch (double amplitude) displacement and of suitable capacity for testing military equipment.

4.6.12.2.3 Test conditions. The test bed of the package tester shall be covered with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. The plywood shall be secured with six-penny nails, with top of heads flush with or slightly below the surface. Nails shall be spaced at 6-inch intervals around all four edges. If the distance between either pair of fences is greater than 24 inches, the plywood shall also be nailed at 3-inch intervals in a 6-inch square at the center of the test area. Using suitable wooden fences, constrain the test item to a horizontal motion of not more than 2 inches in a direction parallel to the axes of the axes of the shafts, a distance more than sufficient to insure the test item will not rebound from fence to fence. For large items, care should be taken to avoid potential air-cushioning effects which may exist between the surface of the package tester and the test item.

4.6.12.2.4 Performance of test. The test item, as secured in its transit case, or combination case, or as otherwise prepared for field transportation, shall be placed on the package tester within the constraints outlined above. The test item will not be operated during vibration. The package tester shall be operated in the synchronous mode with the shafts in phase. (In this mode any point on the bed of the package tester will move in a circular path in a vertical plane perpendicular to the axes of the shafts.) The package tester shall be operated at 1-inch double amplitude and 284 rpm ±2 rpm for a total of 3 hours. At the end of each 1/2-hour period, turn the test item to rest on a different face, so that at the end of the 3-hour period the test item will have rested on each of its six faces (top, bottom, sides, and ends). At the end of the 3-hour period, the test item shall be operated and inspected and results obtained in accordance with General Requirements, 3.2. The package tester shall be operated in the vertical linear mode (straight up and down in the vertical plane) instead of in the synchronous mode when one of the following conditions occurs:

a. Bouncing of the test item is very severe and presents a hazard to personnel.

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b. Forward and rear oscillations cannot be reduced. When operated in the vertical linear mode, wooden fences shall be placed on all four sides of the test item to constrain its motion to not more than 2 inches in either direction.

4.6.13 Procedure XII. For shipboard and amphibious equipment or when a ship is the common carrier, the vibration test shall be in accordance with type I of MIL-STD-167.

4.6.14 Procedure XIII. Transportation vibration test for system, van or shelter assemblage.

4.6.14.1 Part 1. DISCONTINUED

4.6.14.2 Part 2, bounce, system shelter assemblage

4.6.14.2.1 Purpose. To insure that the system shelter assemblage shall be capable of withstanding the vibrations normally induced during transportation. The system shelter assemblage may consist of equipment mounted in a truck or trailer, or equipment mounted in a shelter which is then mounted on a truck or trailer.

4.6.14.2.2 Performance of test. The tire pressure of the van or shelter transport vehicle shall be adjusted for tactical (off-road) cross-country service. The system shelter assemblage shall be driven five times over sections of the Munson Test Course at the Aberdeen Proving Ground, Aberdeen, Maryland, or approved equal, in the following order and at the specified speeds:

Coarse washboard (6-inch waves spaced 72 inches apart)	5 mph
Belgian block	20 mph
Radial washboard (2-inch to 4-inch waves)	15 mph
Two-inch washboard	10 mph
Three-inch spaced bump	20 mph

5. SUMMARY. The following details shall be as specified in the equipment specification or test plan:

a. Equipment category, applicable table, procedure, figure, and test curve (see 3).

b. Pretest data required (General Requirements, 3.2).

c. Failure criteria

d. Weight of test item when acceleration derating of selected curve is required (see 4.5.1 and table 514.2-IIA, note 6).

e. Temperature extremes and temperature test time durations (see 4.3).

f. Total vehicle mileage for category f equipment (see table 514.2-VI).

g. Necessary parameters for calculating test levels for equipment categories 6.2, d.1 and d.2.

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Table 514.2-II. Test Procedure and Time Schedule Chart for Equipment Installed in Propeller Airplanes - Equipment Category b.1

A. Procedure Selection and Time Schedule Chart

Equipment mounting configuration	Procedure number	Procedure part number	Applicable tests (see 4 for test procedures)			Test time schedule (per axis)			Fig. 514.2-2
			Resonance search (4.5.1.1)	Resonance dwell (4.5.1.2)	Sinusoidal cycling (4.5.1.3)	Dwell time at each resonance (4.5.1.2)	Sinusoidal cycling time	Sweep time 5-500-5 Hz / 5-2000-5 Hz	Curve 1/
Without vibration isolators	1	1	X	X	X	30 min	3 hrs less dwell time	15 min / 20 min	C, D, E, F, G, H, J, or L
With vibration isolators 2/	1	1	X	X	X	30 min	3 hrs less dwell time	15 min / 20 min	C, D, E, F, G, J, H, or L
		2	X	X	X	10 min	30 min	15 min / 20 min	B, AR
Normally with vibration isolators but tested without isolators	1	2	X	X	X	30 min	3 hrs less dwell time	15 min / 20 min	B, AR

1/ For sinusoidal vibration resonance tests and cycling tests of item mounted in airplanes and weighing more than 80 pounds, the vibratory accelerations shall be reduced by 1g for each 20-pound increment of weight over 80 pounds. Acceleration derating shall apply only to the highest test level of the selected test curve, however, the vibratory acceleration shall in no case be less than 50 percent of the specified curve level.

2/ Test items of equipment normally provided with vibration isolators first shall be tested with the isolators in place (part 1). The isolators then shall be removed, and test item rigidly mounted and subjected to the test level indicated (part 2).

B. Curve Selection Chart for Category b.1 Equipment

Selection criteria	Fig. 514.2-2 Curve (for freq. to 500 Hz)	Fig. 514.2-2 Curve (for freq. to 2000 Hz for jet engines)
Equipment installed on vibration isolated panels or racks when the panel or rack is not available for test or when the equipment is tested with isolators removed as specified by the applicable procedure.	B	AR
Equipment in forward half of fuselage or equipment in wing areas of airplanes with engines at rear of fuselage.	C	J
Equipment in rear half of fuselage or equipment in wing areas of airplanes with wing or front mounted engines or other equipment of engine locations not specifically mentioned for other curves.	D	H
Equipment located in the engine compartments or engine pylons of airplanes.	E	G
Equipment mounted directly on airplane engines.	F	L

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TABLE 514.2-III
Test procedure and time schedule chart for equipment installed in helicopters-equipment category (c)

Equipment mounting configuration	Procedure number	Procedure part number	Test time schedule (per axis)		Fig. 514.2-3 Curve (note 1)
			Sinusoidal cycling (4.5.1.3)	Sweep Time (Note 3)	
Without vibration isolators	1	1	3 hrs	36 min	M
With vibration isolators (note 2)	1	1	3 hrs	36 min	M
		2	30 min	30 min	B
Normally with vibration isolators but tested without isolators	1	2	30 min	30 min	B

Note 1: For sinusoidal vibration cycling tests of items mounted in helicopters and weighing more than 80 pounds, the vibratory accelerations shall be reduced by 1 g for each 20-pound increment of weight over 80 pounds. Acceleration derating shall apply only to the highest test level of the selected test curve. However, the vibratory acceleration shall in no case be less than 50 percent of the specified curve level.

Note 2: Test items of equipment normally provided with vibration isolators first shall be tested with the isolators in place.

Note 3: Sweep for curve M is 5-2000-5Hz, curve B is 5-500-5 Hz.

B. Curve selection chart for category (c) equipment

Selection criteria	Curve
Equipment designed for installation without vibration isolators	M
Equipment installed on vibration isolated panels or racks when the panel or rack is not available for test or when the equipment is tested with the isolators removed as specified by the applicable procedure.	B

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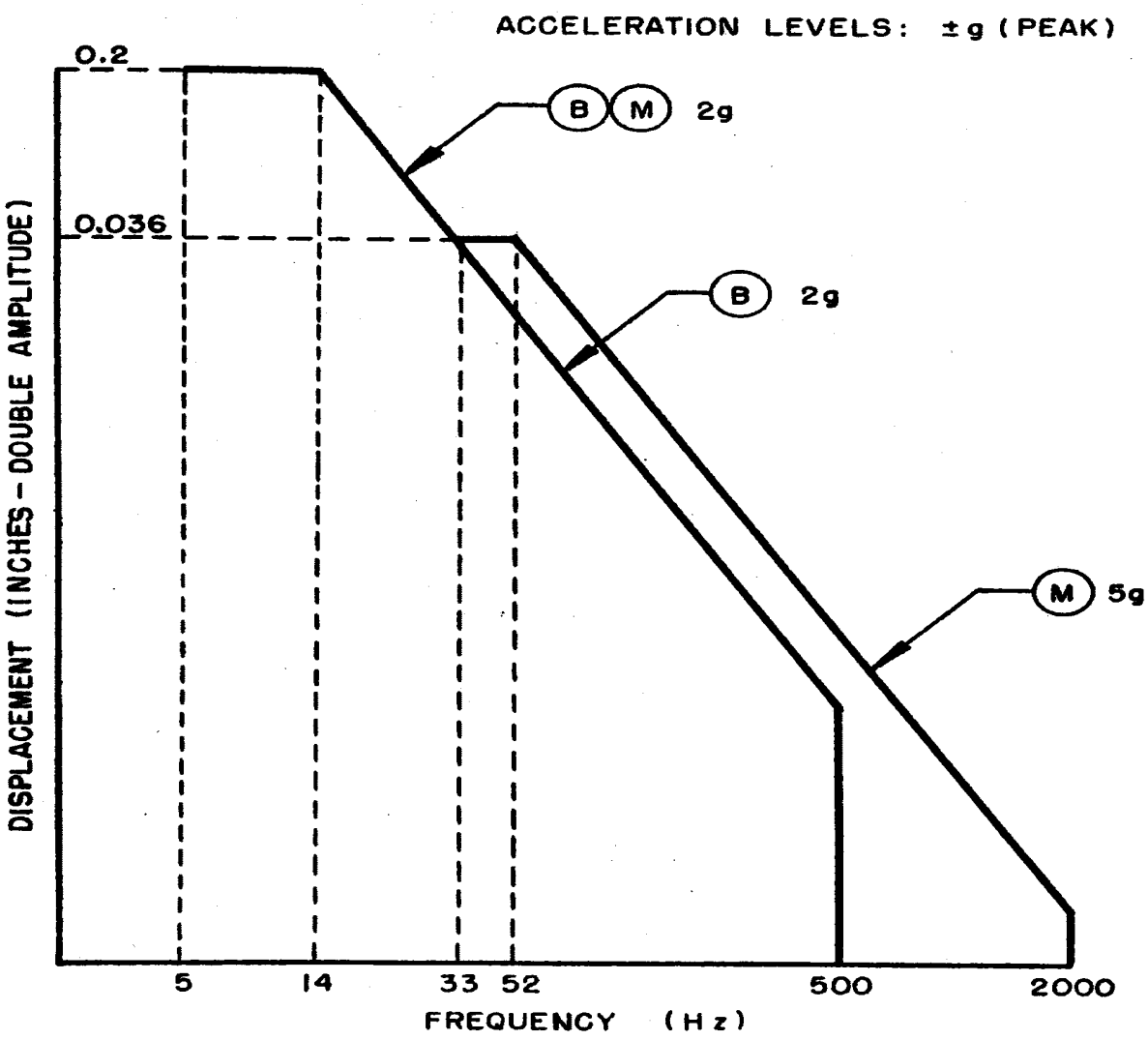


Figure 514.2-3. Vibration Test Curves for Equipment Installed in Helicopters, Equipment Category c.

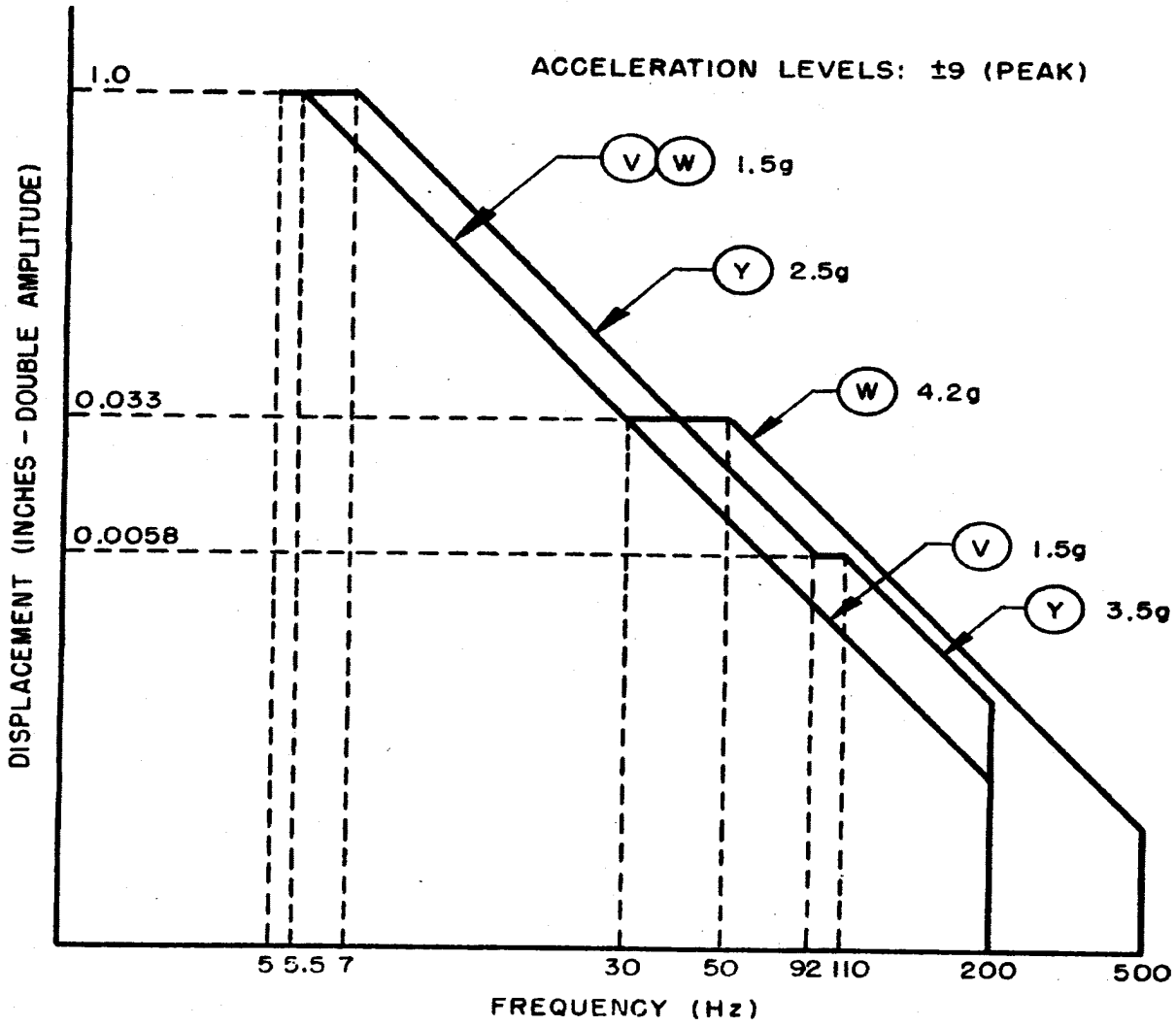
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Table 514.2-VI. Test Procedure and Time Schedule Chart for Equipment Installed in Ground Vehicles, Equipment Category E

Equipment Conditions	Procedure Number	Procedure Part Number	Applicable Tests		Test Time Schedule (Per Axis)			Curve
			Sinusoidal Cycling (4.5.1.3)	Bounce Vehicular (4.6.10.2)	Sinusoidal Cycling Time (4.5.1.3)	Maximum Cycling Time	Sweep Time	
					Schedule A			
Tracked Vehicles	VIII		X		30 min/1000 miles 2/	3 hours	15 min S-500-5 Hz 1/	W
					Schedule B			
Wheeled Vehicles	VIII		X		30 min/1000 miles 2/	5-1/2 hours	12 min S-200-5 Hz 1/	V or Y 3/
Vehicle and Mileage Unknown	VIII		X			3 hours	15 min S-500-5 Hz 1/	W
To be used only when specified	IX	2		X	see 4.6.10.2			
Vans and shelters	XIII	2		see 4.6.14.2				

1/ Sweep time shall be increased by 3 minutes if test frequencies go to 2 Hz.
2/ Cycling time shall be 30 min/1000 miles or as specified in the equipment specification except that it shall not exceed the maximum specified in table 514.2-VI.
3/ Curve V is for equipment installed in wheeled vehicles except for two-wheeled trailers. Curve Y is for equipment installed in two-wheeled trailers.

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Note: All curves shall be extended to 2 Hz when test item resonances below 5 Hz are expected

Figure 514.2-6. Vibration Test Curves for Equipment Installed in Ground Vehicles, Equipment Category f.

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